HARDWARE CALIBRATION STANDARD OPERATING PROCEDURE

AEM13921 APM Calibration Guide

Custom Breakout Procedure Using Nordic PPK2, Owon HDS242 DMM, LD1117 Linear Regulators, and nRF54L15 I^{2}C/BLE Controller

1. Overview & Physical Meaning of Calibration

The e-peas AEM13921 features an internal Average Power Monitoring (APM) block that estimates the electrical power flowing into the storage element (`STO`) from the boost harvester input (`SRC1`) and flowing out of the storage element to supply the regulated application rail (`LOAD`). Because the internal APM state machine measures inductor switch timing rather than physical current or inductance, each PCB specimen requires a one-time room-temperature calibration to determine two conversion constants: $\boldsymbol{\alpha_{\mathbf{SRC1}}}$ and $\boldsymbol{\alpha_{\mathbf{LOAD}}}$.

Note that there is no separate α factor for the `STO` pin itself; all APM measurements are referenced to the power entering or leaving the storage node (`STO`). In Power Meter mode, the AEM13921 compresses the accumulated switching count over each measurement window into a 3-byte floating-point register set (`APM0` , `APM1` , `APM2`) consisting of a mantissa (`power`) and bit shift (`offset`). Unpacking this register via `power << offset` yields a linear digital count proportional to transferred power.

2. Required Equipment & Passive Components

A laboratory 4-quadrant Source Measurement Unit (SMU) can both source and sink current while holding a fixed voltage. Because the Nordic Power Profiler Kit II (PPK2) and LD1117 linear regulators are 1-quadrant devices (they can only source current), calibrating `SRC1` requires a preload resistor (R_{preload}) across the `STO` LDO output so that current pumped out of `STO` by the AEM13921 boost converter is absorbed without raising the `STO` rail voltage.

Component / Tool	Value / Specification	Role in Calibration Setup
Nordic PPK2	Ampere Meter & SMU Modes	Measures time-averaged current ($I_{\text{STO,PPK2}}$) on `STO` ; logic input `D0` captures exact MCU measurement window duration (ΔT).
Owon HDS242	Precision DMM Mode	Measures true Kelvin sense voltage ($V_{\text{STO,DMM}}$) directly at the AEM13921 `STO` and `GND` pins, eliminating PPK2 shunt drop errors.
nRF54L15 Breakout	I^{2}C Master + GPIO + BLE 5	Configures AEM13921 registers, counts `IRQ` rising edges, toggles PPK2 `D0` marker pin, and broadcasts results via BLE Extended Advertising.
USB-PD Triggers (×2)	5 V and 9 V / 12 V DC	Supplies raw DC input power to the two adjustable LD1117 linear regulator stages (`PSU1` for `SRC1` and the clamped source for `STO`).

Component / Tool	Value / Specification	Role in Calibration Setup
LD1117 Adjustable (×2)	1.25 V ref LDOs	Provides stable, low-ripple DC voltages for $V_{\text{STO,CALIB}}$ (on `STO`) and V_{PSU1} (feeding `SRC1` through R_1).
LDO Feedback Network	R_{top} = 100–220 Ω + 1 kΩ Pot	Sets LD1117 output voltage: V_{OUT} = 1.25 V × (1 + R_{pot} / R_{top}). Include 100 nF input and 10 µF electrolytic/low-ESR output capacitors per LDO.
Preload Resistor (R_{preload})	47 Ω to 100 Ω (≥ 0.25 W)	Placed across the `STO` LD1117 output and `GND` (~53 mA draw at 3.6 V with 68 Ω) so the LDO can sink ~2–10 mA coming from `STO` during `SRC1` boost testing.
Source Resistor (R_1)	100 Ω to 1 kΩ (e.g., 150 Ω)	Placed in series between `PSU1` and the AEM13921 `SRC1` pin to emulate harvester impedance and limit input current to 5–15 mA.
Load Resistor (R_2)	500 Ω to 2 kΩ (e.g., 680 Ω)	Placed between the AEM13921 `LOAD` pin and `GND` during buck converter calibration to draw a steady 1–15 mA load current (~5 mA typical).

3. Mathematical Formulas & Variable Definitions

Before wiring the bench, determine the target calibration voltages based on your storage element thresholds (V_{OVDIS} and V_{OVCH}) and harvester Maximum Power Point (MPP) limits ($V_{\text{MPP,MIN}}$ and $V_{\text{MPP,MAX}}$):

1. Storage & Harvester Calibration Voltages:

$V_{\text{STO,CALIB}} = (V_{\text{OVCH}} + V_{\text{OVDIS}}) / 2$ | $V_{\text{SRC1,CALIB}} = (V_{\text{MPP,MAX}} + V_{\text{MPP,MIN}}) / 2$

2. Source Supply Voltage (V_{PSU1}) for ~10 mA Harvester Current:

$V_{\text{PSU1}} = (R_1 \times 0.010\ \text{A}) + V_{\text{SRC1,CALIB}}$ *(Must satisfy: $V_{\text{SRC1,CALIB}} + 0.5\ V \le V_{\text{PSU1}} < V_{\text{STO,CALIB}}$)*

3. Unpacked Register Count & Arithmetic Mean Over N Windows:

APM_i = `(uint64_t)power_i << offset_i` | $APM_{\text{AVG}} = (1 / N) \times \sum_{i=1..N} APM_i$

4. Calibration Factor (α_{SRC1} or α_{LOAD}) [Watts per Count]:

$\alpha = P_{\text{STO,SMU}} / APM_{\text{AVG}} = (I_{\text{STO,PPK2}} \times V_{\text{STO,DMM}}) / APM_{\text{AVG}} = ((Q_{\text{STO,PPK2}} / \Delta T) \times V_{\text{STO,DMM}}) / APM_{\text{AVG}}$

5. True Window Period, Instantaneous Power [W], Energy [J], and Watt-Hours [Wh]:

$T_{\text{window}} = \Delta T / N$ | $P\,[\text{W}] = \alpha \times APM_i$ | $E\,[\text{J}] = P \times T_{\text{window}}$ | $E\,[\text{Wh}] = (\alpha \times T_{\text{window}} / 3600) \times APM_i$

- $V_{\mathrm{STO,DMM}}$ **[V]:** Actual static voltage measured directly across the AEM13921 `STO` and `GND` pins using the Owon HDS242 while the converter is actively running.
- $I_{\mathrm{STO,PPK2}}$ **[A]** & $Q_{\mathrm{STO,PPK2}}$ **[C]:** Average current and total electric charge reported by the PPK2 across the exact time interval (ΔT) where digital marker `D0 == HIGH`.
- APM_{AVG} **[counts]:** Mean of the unpacked `power << offset` values across N consecutive `IRQ` rising edges (typically N = 256 or 258 windows for ~60 seconds at the 233 ms setting).

> **Critical Measurement Rule — Why $V_{\mathrm{STO,DMM}}$ Must Be Measured at the Pin:** The PPK2 measures current using internal switchable series shunt resistors (~10 Ω in the milliamp range). At 3–5 mA, the burden voltage across the PPK2 shunt shifts the actual voltage at the AEM13921 `STO` pin by 30–65 mV relative to the regulator setpoint. Always probe `STO` directly at the breakout header with the Owon HDS242 during active operation and use that reading for $V_{\mathrm{STO,DMM}}$.

4. Procedure A: Calibrating α_{SRC1} (Boost Converter)

During `SRC1` calibration, the harvester boost converter extracts power from `SRC1` and injects current *out* of the `STO` pin. Disconnect all real harvesters, batteries/supercapacitors, and application loads from the breakout board before starting.

4.1 Hardware Wiring Setup

1. **Prepare `STO` Sink/Source Rig:** Adjust LD1117 #1 to $V_{\mathrm{STO,CALIB}}$ (e.g., 3.60 V) and connect R_{preload} (47–100 Ω) directly across LD1117 #1 `VOUT` and `GND`.
2. **Connect PPK2 in Ampere Meter Mode:** Connect PPK2 `VIN` to the AEM13921 `STO` pin and PPK2 `VOUT` to the LD1117 #1 `VOUT` / R_{preload} node (so current flowing out of `STO` registers as positive). Connect common `GND`.
3. **Prepare `SRC1` Supply:** Adjust LD1117 #2 (`PSU1`) to V_{PSU1} (e.g., 3.00 V, or 2.49 V for linearity verification) and connect it through series resistor R_1 (e.g., 150 Ω) to the AEM13921 `SRC1` pin.
4. **Connect Synchronization & Sense Lines:** Wire the AEM13921 `IRQ` pin and I^{2}C lines (`SDA` , `SCL` pulled up to `VINT`) to the nRF54L15. Wire an nRF54L15 marker GPIO to PPK2 digital input `D0`. Clip the Owon HDS242 across the AEM13921 `STO` and `GND` pins.

4.2 Execution & I^{2}C Register Configuration

1. Switch ON the `STO` supply (LD1117 #1) first, then switch ON the `SRC1` supply (LD1117 #2). Wait 2–3 seconds for the AEM13921 to finish coldstart (`VINT` stabilizes at 2.2 V).

2. Configure the AEM13921 registers via I^2C from the nRF54L15:
   - **Boost Settings:** Enable High Power (HP) mode; set the boost timing multiplier to match your target application.
   - **Source Settings:** Set `SRC1` regulation mode to Constant Voltage at $V_{\mathrm{SRC1,CALIB}}$.
   - **Storage Thresholds:** Set V_{OVDIS} ≥ 300 mV below $V_{\mathrm{STO,CALIB}}$, V_{CHRDY} ≥ 200 mV below $V_{\mathrm{STO,CALIB}}$, and V_{OVCH} ≥ 200 mV above $V_{\mathrm{STO,CALIB}}$.
   - **Load Buck:** Disable the `LOAD` buck converter.
   - **APM & Thermal:** Enable APM for `SRC1`, set APM mode to **Power Meter**, set APM window to **233 ms**, set APM accumulator to **1**, and disable thermal protection.
3. Silence the nRF54L15 BLE radio using `bt_le_ext_adv_stop(ext_adv)`.
4. Wait for the next `IRQ` rising edge (Edge 0): clear the interrupt status, discard Edge 0's data, and immediately set GPIO `D0 = HIGH`.
5. For Edges 1 through N (e.g., N = 258, ~61.4 s), read `APM0SRC1` – `APM2SRC1` on each `IRQ` rising edge, unpack `power << offset`, and add to a 64-bit accumulator. Record the stable $V_{\mathrm{STO,DMM}}$ reading from the Owon HDS242.
6. On Edge N, immediately pull GPIO `D0 = LOW`, compute $APM_{\mathrm{SRC1,AVG}}$, and restart BLE advertising with `bt_le_ext_adv_start()` to read the result on your phone.
7. In the nRF Connect Power Profiler app, highlight the exact region where `D0 == HIGH`, record $I_{\mathrm{STO,PPK2}}$ (or $Q_{\mathrm{STO,PPK2}}$ and ΔT), and calculate α_{SRC1} using Formula 4.

5. Procedure B: Calibrating α_{LOAD} (Buck Converter)

During `LOAD` calibration, the internal buck converter draws power *from* `STO` to supply resistor R_2 on the `LOAD` pin. Because `STO` is now purely sourcing current into the AEM13921, you can either keep the LD1117 + PPK2 Ampere Meter rig (reversing PPK2 `VIN` / `VOUT` leads) or use the PPK2 directly in Source Meter (SMU) mode on `STO`.

5.1 Hardware Wiring Setup

1. **Connect Load Resistor (R_2):** Place a 500 Ω to 2 kΩ resistor across the AEM13921 `LOAD` and `GND` pins (selected to draw ~3–5 mA at your target buck voltage: $R_2 \approx V_{\mathrm{LOAD}}$ / 0.005 A).
2. **Supply `STO`:** Connect the PPK2 in Source Meter mode (set to $V_{\mathrm{STO,CALIB}}$, e.g., 3600 mV) or your LD1117 + PPK2 Ampere Meter rig (with PPK2 `VIN` on the LDO side and `VOUT` on the `STO` pin side) to `STO` and `GND`. Keep the Owon HDS242 clipped directly across the AEM13921 `STO` and `GND` pins.

5.2 Execution & I²C Register Configuration

1. Switch ON the `STO` supply. Momentarily apply your `SRC1` supply (for 1–2 seconds) until `VINT` reaches 2.2 V to complete coldstart, then **completely disconnect the `SRC1` wire** so zero power enters from `SRC1`.
2. Configure the AEM13921 registers via I²C from the nRF54L15:
   - **Boost Settings:** Disable both `SRC1` and `SRC2` boost converters.
   - **Storage Thresholds:** Keep V_{OVDIS}, V_{CHRDY}, and V_{OVCH} configured around $V_{\mathrm{STO,CALIB}}$.
   - **Load Buck:** Enable the `LOAD` buck converter; set the buck timing multiplier and target `LOAD` output voltage according to your application requirements.
   - **APM & Thermal:** Enable APM for `LOAD`, set APM mode to **Power Meter**, set APM window to **233 ms**, set APM accumulator to **1**, and disable thermal protection.
3. Pause BLE advertising, execute the exact same N = 258 window synchronized capture using `IRQ` and GPIO `D0`, record $V_{\mathrm{STO,DMM}}$ on the Owon HDS242 while the buck converter is active, and calculate α_{LOAD} using Formula 4.

6. Firmware Implementation Pattern (nRF54L15 / Zephyr)

To prevent 32-bit floating-point precision loss over long deployments, accumulate the unpacked integer counts (`power << offset`) in a 64-bit unsigned integer on every `IRQ` rising edge, and apply the floating-point scaling factors only when formatting telemetry. During calibration, stop extended advertising with `bt_le_ext_adv_stop(ext_adv)` rather than calling `bt_disable()` so the BLE radio stays silent without invalidating the advertising handle.

```
// Pre-calculated constants for Board #1 (T_window = 0.23798 s)
constexpr double ALPHA_SRC1_W     = 6.36e-9;   // Watts per count
constexpr double ALPHA_LOAD_W     = 5.76e-9;   // Watts per count
constexpr double WH_PER_CNT_SRC1  = 4.204e-13; // Watt-hours per count (ALPHA_SRC1
* 0.23798 / 3600)
constexpr double WH_PER_CNT_LOAD  = 3.808e-13; // Watt-hours per count (ALPHA_LOAD
* 0.23798 / 3600)

static uint64_t total_src1_counts = 0;
static uint64_t total_load_counts = 0;

void on_aem_irq_rising_edge(uint16_t src1_pwr, uint8_t src1_ofs, uint16_t
load_pwr, uint8_t load_ofs) {
    total_src1_counts += (static_cast<uint64_t>(src1_pwr) << src1_ofs);
    total_load_counts += (static_cast<uint64_t>(load_pwr) << load_ofs);
}
```

7. Reference Calibration Log (Breakout Board Specimen #1)

The table below records the verified calibration measurements obtained on your first custom AEM13921 breakout board (N = 258 windows, nominal 233 ms setting, actual measured duration ΔT = 61.4 s → true window period $\boldsymbol{T_{\mathbf{window}}}$ **= 237.98 ms**):

Test Run	$V_{\mathrm{STO,DMM}}$	PPK2 Avg / Max I_{STO}	PPK2 Charge (Q)	Mean Count (APM_{AVG})	Calibrated α [W/cnt]	Energy Factor [Wh/cnt]
SRC1 Run 1 (V_{PSU1} ≈ 3.00 V)	3.561 V	3.95 mA / 4.84 mA	242.67 mC (61.4 s)	2,212,804.00	**6.357×10^{-9}**	4.202×10^{-13}
SRC1 Run 2 (V_{PSU1} ≈ 2.49 V)	~3.561 V	2.58 mA / 4.42 mA	158.41 mC (61.4 s)	1,435,816.19	**6.399×10^{-9}**	4.230×10^{-13}
LOAD Run 1 (Buck Active)	3.542 V	3.46 mA / 19.09 mA	212.71 mC (61.4 s)	2,127,173.66	**5.761×10^{-9}**	3.808×10^{-13}